

VIRTUAL UNIVERSITY OF PAKISTAN

A Good Education is a Foundation For a Better Future

CORRECT ANSWER SOLVED BY HADI
EMAIL: hadirajputofficial@gmail.com
WHATSAPP& CALL: 03087122922

TEAM HADI(VU LAHORE)

MID TERM PAST PAPER

We Deal All Kind of VU Project
(PHP, .NET, ANDROID, PYTHON)

MADE AND SOLVED BY HADI

WARNING: Team HADI is not responsible for any mistake or wrong answer. All students reading and using this document may check and confirm the answers at their own.



WhatsApp: 03087122922



FACEBOOK ID: <https://www.facebook.com/hadipastpapers/>

Best of luck!



Question No : 1 of 26

Marks: 1 (Budgeted Time 1 Min)

$7x$ is an algebraic term in which 7 is a _____ and x is a _____.

Answer (Please select your correct option)

☐ term, expression

☐ coefficient, variable

☐ variable, coefficient

☐ numerical, alphabet



A Good Education is a Foundation For a Better Future

CORRECT ANSWER SOLVED BY HADI

EMAIL: hadirajputofficial@gmail.com

WHATSAPP& CALL: 03087122922

MADE BY HADI

Email: hadirajputofficial@gmail.com Whatsapp: 03087122922 (Team Hadi Lahore)

Question No : 2 of 26

Marks: 1 (Budgeted Time 1 Min)

Which of the following is the pivot element in the second row of the matrix?

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 8 \\ 0 & 0 & -3 & 5 & 1 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & -3 \end{pmatrix}$$

Answer (Please select your correct option)

0

☐

1

☐

-3

☒

5

☐



A Good Education is a Foundation For a Better Future

CORRECT ANSWER SOLVED BY HADI

EMAIL: hadirajputofficial@gmail.com

WHATSAPP& CALL: 03087122922

MADE BY HADI

Question No : 3 of 26

Marks: 1 (Budgeted Time 1 Min)

Which of the following is true for the matrix $\begin{pmatrix} 1 & 3 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$?

Answer (Please select your correct option)

☐ It is an identity matrix.

☐ It is in reduced echelon form.

☒ It is in echelon form.

☐ It is a rectangular matrix.



A Good Education is a Foundation For a Better Future

CORRECT ANSWER SOLVED BY HADI

EMAIL: hadirajputofficial@gmail.com

WHATSAPP & CALL: 03087122922

MADE BY HADI

Question No : 4 of 26

Marks: 1 (Budgeted Time 1 Min)

If reduced echelon form of a linear system is $\begin{bmatrix} 1 & 0 & 5 & 5 \\ 0 & 1 & 1 & 6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ when free variable $x_3 = 0$, then which of the following is true for it?

Answer (Please select your correct option)

- ☐ The particular solution is $(0, 5, 6)$.
- ☐ The particular solution is $(6, 5, 0)$.
- ☐ The particular solution is $(5, 6, 0)$.
- ☐ The particular solution is $(0, 6, 5)$.



A Good Education is a Foundation For a Better Future

CORRECT ANSWER SOLVED BY HADI
EMAIL: hadirajputofficial@gmail.com

MADE BY HADI

Question No : 5 of 26

Marks: 1 (Budgeted Time 1 Min)

If $\vec{b} = c_1 \vec{u}_1 + c_2 \vec{u}_2$ and $\vec{a} = d_1 \vec{u}_1 + d_2 \vec{u}_2$ where c_1, c_2, d_1 and d_2 are scalars, then which of the following options is correct?

Answer (Please select your correct option)

☐ Only $\vec{b} \in \text{Span}\{\vec{u}_1, \vec{u}_2\}$.

☐ Only $\vec{a} \in \text{Span}\{\vec{u}_1, \vec{u}_2\}$.

☐ Both $\vec{a}, \vec{b} \in \text{Span}\{\vec{u}_1, \vec{u}_2\}$.

☐ Both $\vec{a}, \vec{b} \notin \text{Span}\{\vec{u}_1, \vec{u}_2\}$.



A Good Education is a Foundation For a Better Future

CORRECT ANSWER SOLVED BY HADI
EMAIL: hadirajputofficial@gmail.com
WHATSAPP & CALL: 03087122922

MADE BY HADI

Email: hadirajputofficial@gmail.com | WhatsApp & Call: 03087122922 | Team Had

Question No : 6 of 26

Marks: 1 (Budgeted Time 1 Min)

If a homogeneous system $Ax = 0$ has a trivial solution, then which of the following is (are) the value(s) of the vector x ?

Answer (Please select your correct option)

-1

☐

0

☐

1

☐

2

☐

A Good Education is a Foundation For a Better Future

CORRECT ANSWER SOLVED BY HADI
EMAIL: hadirajputofficial@gmail.com
WHATSAPP& CALL: 03087122922

MADE BY HADI

Email: hadirajputofficial@gmail.com Whatsapp: 03087122922 (Team Hadi Lahore)

Question No : 7 of 26

Marks: 1 (Budgeted Time 1 Min)

If a homogeneous system $AX = 0$ has non-trivial solution, then which of the following is true for the system?

Answer (Please select your correct option)

- ☐ The system has at least no free variable.
- ☐ The system has at least one free variable.
- ☐ The system has at least two free variables.
- ☐ The system has at least three free variables.

A Good Education is a Foundation For a Better Future

CORRECT ANSWER SOLVED BY HADI
EMAIL: hadirajputofficial@gmail.com
WHATSAPP & CALL: 03087122922

MADE BY HADI

Email: hadirajputofficial@gmail.com Whatsapp: 03087122922 (Team Hadi Lahore)

Question No : 8 of 26

Marks: 1 (Budgeted Time 1 Min)

If $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1+6x_3 \\ 3 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 6 \\ 0 \\ 1 \end{bmatrix}$ be the general solution of $Ax = b$, then which of the following is parametric equation for the given solution?

Answer (Please select your correct option)

☐ $p = sx + tu + v$

☐ $x = p + tv$

☐ $v = p + tx$

☐ $tx = p + v$



A Good Education is a Foundation For a Better Future

CORRECT ANSWER SOLVED BY HADI
EMAIL: hadirajputofficial@gmail.com
WHATSAPP & CALL: 03087122922

MADE BY HADI

Email: hadirajputofficial@gmail.com Whatsapp: 03087122922 (Team Hadi Lahore)

Question No : 9 of 26

Marks: 1 (Budgeted Time 1 Min)

If the equation $T(\vec{x}) = \vec{0}$ has only the trivial solution, then which of the following is true for the linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$?

Answer (Please select your correct option)

☐ T is one-to-one.



☐ T is onto.

☐ T is a rotation.

☐ T is a reflection.

A Good Education is a Foundation For a Better Future

CORRECT ANSWER SOLVED BY HADI
EMAIL: hadirajputofficial@gmail.com
WHATSAPP & CALL: 03087122922

MADE BY HADI

Email: hadirajputofficial@gmail.com Whatsapp: 03087122922 (Team Hadi Lahore)

Question No : 10 of 26

Marks: 1 (Budgeted Time 1 Min)

If linear transformation $T : R^n \rightarrow R^m$ is transformed into a matrix A , then which of the following is the order of A ?

Answer (Please select your correct option)

☐ $n \times m$

☐ $m \times m$

☒ $m \times n$

☐ $n \times n$



A Good Education is a Foundation For a Better Future

CORRECT ANSWER SOLVED BY HADI

EMAIL: hadirajputofficial@gmail.com

WHATSAPP & CALL: 03087122922

MADE BY HADI

If $A = A^t$ (where A is a square matrix) , then which of the following is the most appropriate option for A ?

Answer (Please select your correct option)

☐ A is an invertible matrix.

☐ A is a singular matrix.

☐ A is a symmetric matrix.

☐ A is a scalar matrix.



A Good Education is a Foundation For a Better Future

CORRECT ANSWER SOLVED BY HADI
EMAIL: hadirajputofficial@gmail.com
WHATSAPP & CALL-03087122922

MADE BY HADI

Question No : 12 of 26

Marks: 1 (Budgeted Time 1 Min)

If A is an invertible matrix , then which of the following is true ?

Answer (Please select your correct option)

☐ $(A^{-1})^{-1} = A$



☐ $(A^{-1})^{-1} = \frac{1}{A}$

☐ $(A^{-1})^{-1} = A^{-1}$

☐ $(A^{-1})^{-1} = \det(A)$

A Good Education is a Foundation For a Better Future

CORRECT ANSWER SOLVED BY HADI
EMAIL: hadirajputofficial@gmail.com
WHATSAPP& CALL: 03087122922

MADE BY HADI

Email: hadirajputofficial@gmail.com Whatsapp: 03087122922 (Team Hadi Lahore)

Question No : 13 of 26

Marks: 1 (Budgeted Time 1 Min)

If $A = \begin{bmatrix} 4 & -1 \\ 5 & 3 \end{bmatrix}$, then which of the following is the value of $\det(A)$?

Answer (Please select your correct option)

☐ 7

☐ -17

☐ 17

☐ 11



A Good Education is a Foundation For a Better Future

CORRECT ANSWER SOLVED BY HADI
EMAIL: hadirajputofficial@gmail.com

MADE BY HADI

Question No : 14 of 26

Marks: 1 (Budgeted Time 1 Min)

If a system of equations is solved using the Jacobi's method , then which of the following is NOT true about the matrix M that is derived from the coefficient matrix ?

Answer (Please select your correct option)

☐ All of its entries below the diagonal must be zero .

☐ All of its entries above the diagonal must be zero .

☐ It may or may not be invertible .

☐ It is a non-singular matrix .



A Good Education is a Foundation For a Better Future

CORRECT ANSWER SOLVED BY HADI

EMAIL: hadirajputofficial@gmail.com

WHATSAPP& CALL :03087122922

MADE BY HADI

Question No : 15 of 26

Marks: 1 (Budgeted Time 1 Min)

If the matrix $A = \begin{bmatrix} 5 & 4 \\ 3 & 3 \end{bmatrix}$, then which of the following is the most suitable option for it ?

Answer (Please select your correct option)

☐ It is not strictly diagonally dominant.

☐ It is strictly diagonally dominant.

☐ It is not diagonally dominant.

☐ It is diagonally dominant.



A Good Education is a Foundation For a Better Future

CORRECT ANSWER SOLVED BY HADI
EMAIL: hadirajputofficial@gmail.com
WHATSAPP& CALL: 03087122922

MADE BY HADI

Email: hadirajputofficial@gmail.com Whatsapp: 03087122922 (Team Hadi Lahore)

Question No : 16 of 26

Marks: 1 (Budgeted Time 1 Min)

If $A = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$, then which of the following is the minor of entry a_{21} ?

Answer (Please select your correct option)

☐ 2

☐ 3

☐ 4

☐ 5



A Good Education is a Foundation For a Better Future

CORRECT ANSWER SOLVED BY HADI
EMAIL: hadirajputofficial@gmail.com
WHATSAPP& CALL: 03087122922

MADE BY HADI

Email: hadirajputofficial@gmail.com Whatsapp: 03087122922 (Team Hadi Lahore)

Question No : 17 of 26

Marks: 1 (Budgeted Time 1 Min)

If $A = \begin{bmatrix} 4 & 5 \\ 6 & 7 \end{bmatrix}$, then which of the following is the value of C_{21} ?

Answer (Please select your correct option)

☐ 5

☒ -5

☐ 6

☐ -6



A Good Education is a Foundation For a Better Future

CORRECT ANSWER SOLVED BY HADI
EMAIL: hadirajputofficial@gmail.com
WHATSAPP & CALL: 03087122922

MADE BY HADI

Email: hadirajputofficial@gmail.com Whatsapp: 03087122922 (Team Hadi Lahore)

If the determinant of the matrix $A = \begin{bmatrix} 2 & 3 & 4 \\ 1 & 2 & 2 \\ 3 & 4 & 5 \end{bmatrix}$ is -1 and the matrix B is obtained by adding 2 times of the second row in the first row of the matrix A , then which of the following is true about the matrix B ?

Answer (Please select your correct option)

- ☐ Its determinant is -1 .
- ☐ Its determinant is 1 .
- ☐ Its determinant can not be evaluated .
- ☐ The information is not sufficient to calculate the determinant .



A Good Education is a Foundation For a Better Future

CORRECT ANSWER SOLVED BY HADI
EMAIL: hadirajputofficial@gmail.com
WHATSAPP & CALL: 03087122922

MADE BY HADI

Email: hadirajputofficial@gmail.com Whatsapp: 03087122922 (Team Hadi Lahore)

Question No : 19 of 26

Marks: 1 (Budgeted Time 1 Min)

If the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix}$, then which of the following is true about it ?

Answer (Please select your correct option)

☐ Its determinant is 0 .

☐ Its determinant is 1 .

☐ Its determinant is 2 .

☐ Its determinant is 4 .

A Good Education is a Foundation For a Better Future

CORRECT ANSWER SOLVED BY HADI

EMAIL: hadirajputofficial@gmail.com

WHATSAPP: CALL - 02087199099

MADE BY HADI

Question No : 20 of 26

Marks: 1 (Budgeted Time 1 Min)

If a set $S = \{ v_1 = (1, 2), v_2 = (4, 8) \}$, then which of the following is the most appropriate option ?

Answer (Please select your correct option)

☐ It is a basis of $\mathbb{R}^2$.

☐ It is linearly independent.

☐ It spans $\mathbb{R}^2$.

☐ It is linearly dependent.



A Good Education is a Foundation For a Better Future

CORRECT ANSWER SOLVED BY HADI
EMAIL: hadirajputofficial@gmail.com
WHATSAPP& CALL: 03087122922

MADE BY HADI

Question No : 21 of 26

Marks: 2 (Budgeted Time 4 Min)

If $A = \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix}$ and $A^{-1} = \begin{bmatrix} 7 & -2 \\ -3 & 1 \end{bmatrix}$, then calculate the inverse of A^t (if exists).

Answer (Please [click here](#) to Add Answer)



A Good Education is a Foundation For a Better Future

CORRECT ANSWER SOLVED BY HADI
EMAIL: hadirajputofficial@gmail.com
WHATSAPP& CALL: 03087122922

MADE BY HADI

Email: hadirajputofficial@gmail.com Whatsapp: 03087122922 (Team Hadi Lahore)

Question No : 22 of 26

Marks: 2 (Budgeted Time 4 Min)

Evaluate the determinant

$$\begin{vmatrix} 2 & 4 & 8 & 16 \\ 0 & 3 & 6 & 12 \\ 0 & 0 & -2 & -4 \\ 0 & 0 & 0 & -3 \end{vmatrix}.$$

Answer (Please [click here](#) to Add Answer)

A Good Education is a Foundation For a Better Future

CORRECT ANSWER SOLVED BY HADI
EMAIL: hadirajputofficial@gmail.com
WHATSAPP& CALL: 03087122922

MADE BY HADI

Email: hadirajputofficial@gmail.com Whatsapp: 03087122922 (Team Hadi Lahore)

Question No : 23 of 26

Marks: 3 (Budgeted Time 6 Min)

If $A = \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 5 \\ 3 & 4 \end{bmatrix}$, then calculate the inverse of AB (if exists); where $A^{-1} = \begin{bmatrix} 7 & -2 \\ -3 & 1 \end{bmatrix}$ and $B^{-1} = \begin{bmatrix} 4 & -5 \\ -3 & 4 \end{bmatrix}$.

Answer (Please [click here](#) to Add Answer)

A Good Education is a Foundation For a Better Future

CORRECT ANSWER SOLVED BY HADI
EMAIL: hadirajputofficial@gmail.com
WHATSAPP& CALL: 03087122922

MADE BY HADI

Email: hadirajputofficial@gmail.com Whatsapp: 03087122922 (Team Hadi Lahore)

Let S be a parallelogram determined by the vectors $\vec{b}_1 = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$, $\vec{b}_2 = \begin{bmatrix} -2 \\ 5 \end{bmatrix}$ and $A_1 = \begin{bmatrix} 6 & -2 \\ -3 & 2 \end{bmatrix}$. Compute the area of the image of S under the mapping $\vec{x} \rightarrow A_1 \vec{x}$.

Answer (Please [click here](#) to Add Answer)



A Good Education is a Foundation For a Better Future

CORRECT ANSWER SOLVED BY HADI
EMAIL: hadirajputofficial@gmail.com
WHATSAPP& CALL-03087122922

MADE BY HADI

Email: hadirajputofficial@gmail.com Whatsapp: 03087122922 (Team Hadi Lahore)

Question No : 25 of 26

Marks: 5 (Budgeted Time 10 Min)

Find an LU - decomposition of the matrix $A = \begin{bmatrix} 2 & 3 \\ 5 & 7 \end{bmatrix}$.

Answer (Please [click here](#) to Add Answer)



A Good Education is a Foundation For a Better Future

CORRECT ANSWER SOLVED BY HADI
EMAIL: hadirajputofficial@gmail.com
WHATSAPP& CALL: 03087122922

MADE BY HADI

Email: hadirajputofficial@gmail.com Whatsapp: 03087122922 (Team Hadi Lahore)

If $A = \begin{bmatrix} 4 & 1 & 6 \\ -7 & 5 & -3 \\ 9 & -3 & 3 \end{bmatrix}$, then find the minors of first row and determinant of A using these minors.

Answer (Please [click here](#) to Add Answer)



A Good Education is a Foundation For a Better Future

CORRECT ANSWER SOLVED BY HADI
EMAIL: hadirajputofficial@gmail.com
WHATSAPP& CALL: 03087122922

MADE BY HADI

Email: hadirajputofficial@gmail.com Whatsapp: 03087122922 (Team Hadi Lahore)

HADI VU PAST PAPER

WE DEAL ALL KIND OF VU PROJECT
(PHP, ANDROID, PYTHON)

DEAR STUDENTS

ARE YOU LOOKING
FOR ASSISTANCE

- 1) ASSIGNMENTS
- 2) QUIZZES
- 3) GDB
- 4) GRAND QUIZZES

We Provide, You Solution
of VU Academic Activities
include Assignments,
Quizzes, GDBs, Grand Quizzes
If You Need Help in Services
Kindly inform us.

TEAM HADI
(VU LAHORE)



TEAM HADI
(VU LAHORE)



HADI TECH

#PAID_TASK

CONTACT US

CALL & WHATSAPP

0308-7122922